

What sets the PING rate: the per-synapse weight, not the pool size

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In a PING (pyramidal-interneuron network gamma) circuit, excitatory cells drive a shared inhibitory pool, and that inhibition feeds back to shunt the excitatory cells. A natural question when scaling such a network is whether the **number** of inhibitory neurons N_I sets the operating rate, or whether it is the **strength** of each inhibitory synapse W^{IE} .

Here the two are swept independently on an **untrained** PING network at the canonical biophysical initialisation — no learning, just the fixed-point rate the loop settles into under uniform Poisson drive. Each excitatory cell receives inhibition

$$I_i^{\text{inh}(t)} = \sum_{j \in I, \text{ spiked}} W_{ij}^{IE} g_i(t),$$

where W^{IE} is the per-synapse weight (mean swept) and g_i is the fixed inhibitory release conductance. The claim is that the rate is a function of the per-event shunt W^{IE} relative to the fixed release level — not of N_I .

The sweep confirms it. Plotting the per-cell E and I rate against W^{IE} , one line per pool size, the $N_I \in \{16, 64, 256\}$ curves land on top of one another: pool size does not move the rate, the per-synapse weight does. Increasing W^{IE} shunts each excitatory cell harder per inhibitory spike, so the whole loop settles at a lower rate — monotonically, across three decades of N_I .

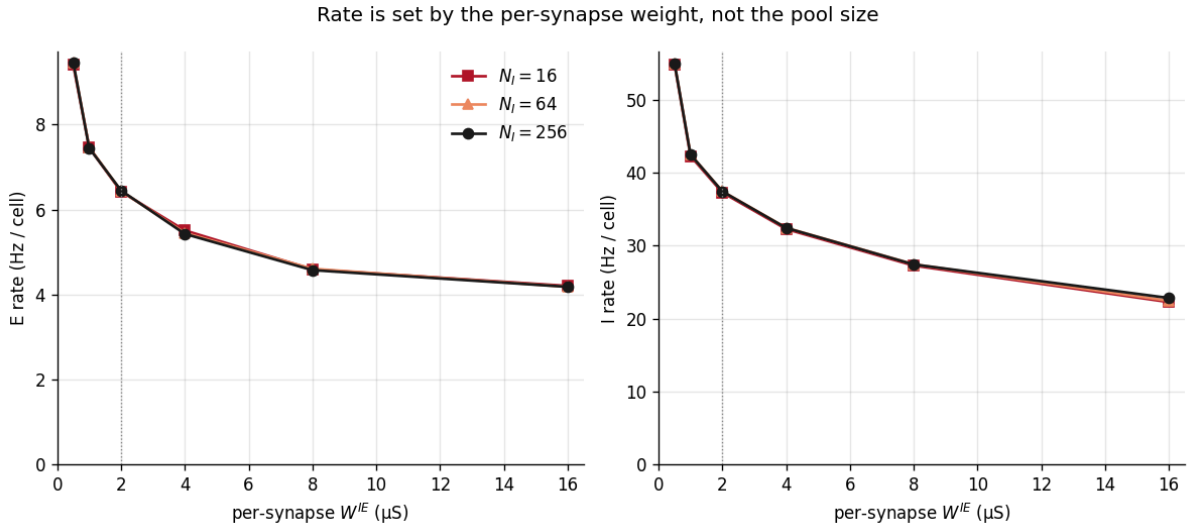


Figure 2: E (left) and I (right) per-cell firing rate versus the per-synapse weight W^{IE} , one line per inhibitory pool size N_I . The lines overlap — the rate is set by W^{IE} , not N_I . The dotted line marks the biophysical-init default.

The three commonly quoted “scalings” — constant per-synapse, synaptic $1/N$, and critical $1/\sqrt{N}$ — are then just rules for **choosing** W^{IE} given N_I , i.e. paths across this single master curve. They need an arbitrary anchor and add nothing to the mechanism, so they are not plotted.

Run scale: 1024 excitatory cells, 4 trials, 200 ms at 0.1 ms steps, uniform Poisson input at 25 Hz, seed 42 — a 6×3 grid of $W^{IE} \times N_I$.